

# Samaira Rana

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## PROFILE

Bachelor's student in Digital Business and Data Science in Germany, currently in the 4th semester, seeking a 5th-semester Praxissemester (Aug 2026 – ). Hands-on experience building and deploying full-stack applications with Python/Flask backends, PostgreSQL, and rule-based AI logic, as part of team and independent projects.

## EDUCATION

**University of Europe for Applied Sciences, B.Sc. Digital Business and Data Science** 03/2025 – 2028 (expected)

Potsdam, Germany

- Completed 3 semesters; core coursework in data science, business informatics, statistics, and programming
- Currently seeking a mandatory Praxissemester (5th semester) starting around March 2027

**Ryan International School — Senior Secondary (Commerce), Score: 84.6%**

– 2024

Patiala, India

## PROJECTS

**JARVIS — Local AI Personal Assistant | Independent Project**

09/2025 – Present

*Independent project · Python, Ollama (Llama 3), faster-whisper, openWakeWord, PyQt6, ChromaDB*

- Built a privacy-first, fully local AI assistant for Windows: wake-word voice control, an on-device LLM for reasoning, a desktop HUD dashboard, and 40+ built-in tools spanning calendar, budget, study, and daily life, with no cloud AI subscription and no data leaving the machine by default
- Implemented always-listening wake-word detection (openWakeWord) with echo suppression while the assistant is speaking and a cooldown window to prevent false retriggers, feeding voice-activity-gated speech-to-text (faster-whisper) that skips silence entirely
- Ran reasoning on a locally-hosted Llama 3 model via Ollama with automatic crash detection, process restart, and retry, plus automatic summarisation of older conversation turns when history approaches the model's context window so nothing is silently truncated
- Built bilingual English/German support that detects the spoken language per utterance and switches the assistant's voice and responses mid-conversation; designed a plugin architecture for adding new tools without touching core logic; wrote a 12-file pytest suite covering budget, closet, alarms, notes, Pomodoro, memory, and integration flows

**Predicting Video Game User Ratings | Independent Machine Learning Project**

08/2026

*Independent project · Python, pandas, scikit-learn (Logistic Regression, Random Forest)*

- Built a machine learning pipeline predicting whether a video game earns a good user rating ( $\geq 7/10$ ) from a real-world Kaggle dataset of ~16,700 games, emphasizing rigorous data cleaning and honest evaluation over headline accuracy
- Uncovered a hidden data-quality issue where ~5% of the target column was stored as the placeholder string "tbd" rather than a true null; exposed it with pandas to\_numeric error coercion and excluded Global\_Sales from the feature set since it is the arithmetic sum of the regional sales columns
- Trained and compared Logistic Regression and Random Forest classifiers on a 3-feature set (Genre, Platform, Critic\_Score) and an extended 9-feature set (+ Publisher, Year, regional sales); all models beat the 63.7% majority-class baseline
- Investigated a 5.3-point accuracy gap between the two models via feature-importance analysis (Critic\_Score drove 72.9% of Random Forest's decisions) and a scatter-plot check of linearity, formed a testable hypothesis, and confirmed it: expanding to 9 features narrowed the gap to 0.4 points

**IDMS — Integrated Data Management System | Scrum Master & DevOps Lead**

Apr 2026 – Jun 2026

*University Agile Development course · [github.com/SamairaRana30/idms-backend](https://github.com/SamairaRana30/idms-backend) · Live: [idms-backend-deu6.onrender.com](https://idms-backend-deu6.onrender.com)*

- Led a 9-person Agile team as Scrum Master across 4 two-week sprints, running sprint planning, daily standups, reviews, and retrospectives; tracked 453 Jira tickets (52 stories, 377 subtasks) end-to-end
- Achieved 100% sprint velocity in 3 of 4 sprints; drove recovery after a sprint dropped to 65% (11/17 stories) when a teammate went unresponsive, by personally implementing the abandoned documents module (upload, secure storage, search, access control) and reassigning the remaining tickets to restore full velocity
- Delivered a full-stack platform (member management, e-voting, meetings, finance tracking, notifications, WhatsApp analytics) to production, migrating ~4,000 member records

- Owned DevOps and contributed to UI/UX: Flask backend (Blueprint factory pattern) with JWT + bcrypt auth, Supabase PostgreSQL across dev/staging/prod, Bootstrap 5 frontend, and continuous deployment to Render
- Oversaw QA covering 47 test cases and 12 bugs, all resolved before the production release

### **Vogueabulary — AI-Based Smart Styling System | AI System Developer (team of 4)** 05/2026 – 07/2026

Team project (4-person team) · Team repo: [github.com/janvisanger22-dotcom/VOCABBULARY](https://github.com/janvisanger22-dotcom/VOCABBULARY) · Live: [vogueabulary.onrender.com](https://vogueabulary.onrender.com)

- Designed and implemented a rule-based AI recommendation engine (Python) that generates ranked outfit combinations from a user's digital wardrobe, filtering by weather and occasion and scoring colour harmony across top, bottom, shoes, and outerwear pairings
- Built a weighted colour-compatibility scoring model grouping garments into tone families (neutral, warm, cool, earth) with pairwise harmony scores driving a 0–100 outfit score
- Integrated an AI styling chat assistant (Groq/Llama 3.3) for personalized, wardrobe-aware advice, plus an LLM-powered outfit verdict (Anthropic Claude) with a rule-based fallback for reliability
- Contributed to Flask backend logic and Google OAuth login integration; helped ship a live, publicly deployed application on Render

### **iPhone YouTube Comment Sentiment Analysis | Data Cleaning & Sentiment Analysis** 03/2025 – 07/2025

Big Data & Analytics course project, team of 1st-semester students · Python, pandas, VADER, Hadoop Streaming

- Owned data cleaning and sentiment analysis within a team project analyzing 20,234 real YouTube comments from iPhone review/launch videos, following a Hadoop → PostgreSQL → Python pipeline brief
- Built a custom Hadoop Streaming MapReduce job (Python mapper/reducer) to clean and normalize the raw comment corpus, stripping URLs, non-ASCII characters, and stopwords
- Performed sentiment analysis with VADER, classifying comments as positive/neutral/negative (48% / 29% / 23%), and found that negative sentiment share rose from 18% to 25% between the iPhone 15 and iPhone 16 launch-video comment spikes
- Built an interactive HTML dashboard to visualize sentiment distribution, monthly trends, engagement, and distinctive vocabulary by sentiment class

### **The Impact of Social Media on the Mental Health of University Students**

03/2025 – 07/2025

Qualitative Research — Data Analysis Lead · University of Europe for Applied Sciences · Group research project (4-person team)

- Contributed to a 4-person group research project exploring how social media use affects the mental health of university students, using a qualitative, phenomenological research design; collected data through semi-structured interviews with 10–15 university students plus a supplementary Google Form questionnaire
- Led the data analysis: applied thematic analysis to the interview and questionnaire data and identified seven recurring themes, including social comparison and inadequacy, compulsive checking and anxiety, filtered self and body image struggles, cyberbullying, escapism and emotional dependence, and peer support and community
- Synthesized the findings into a written report highlighting both the protective and harmful patterns in students' social media use, and contributed recommendations for student mental health and digital wellbeing support

## **TECHNICAL SKILLS**

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- Languages & frameworks: Python, Flask, JavaScript, HTML/CSS, SQL
- Data & AI: pandas, scikit-learn (Logistic Regression, Random Forest), local LLM integration (Ollama/Llama 3), speech-to-text/wake-word pipelines (faster-whisper, openWakeWord), sentiment analysis (TextBlob, VADER), rule-based recommendation systems
- Databases & infra: PostgreSQL (Supabase), Git/GitHub, Docker, Render deployment
- Ways of working: Agile/Scrum (sprint-based delivery), team collaboration on shared codebases
- MS Office: Word, Excel, PowerPoint, Teams

## **CERTIFICATIONS**

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### **EXIN BCS Artificial Intelligence Essentials | EXIN**

Issued Oct 2024

Credential ID 6601761

## **LANGUAGES**

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- English — IELTS 8/9 (C1/C2)
- German — A2
- Hindi — Native

**VOLUNTEER EXPERIENCE**

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**Model United Nations (MUN) | Delegate**

Mar 2022 – Mar 2024

*Ryan International School, Patiala*

- Represented France and Georgia as a delegate in Model United Nations conferences during senior secondary school
- Researched country positions on international issues, drafted formal position papers and resolutions, and prepared and delivered speeches in committee debates — recognized for the thoroughness and depth of research
- Strengthened public speaking, negotiation, and analytical thinking skills through structured diplomatic simulation

**NGO VONG | Volunteer**

Mar 2022 – Mar 2024

*Ryan International School, Patiala*

- Participated in donation drives and community engagement projects supporting local social causes